

Practice Problems #8

1. Let A be a 7×5 matrix. What must m and n be in order to define $T : \mathbf{R}^m \rightarrow \mathbf{R}^n$ by $T(x) = Ax$?
2. With T defined by $T(x) = Ax$, find an x whose image under T is b . Is x unique?

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 3 & 1 & -5 \\ 4 & 2 & 1 \end{bmatrix}, b = \begin{bmatrix} 0 \\ -5 \\ -6 \end{bmatrix}.$$

3. Let $b = \begin{bmatrix} 1 \\ -1 \\ 7 \end{bmatrix}$ and let $A = \begin{bmatrix} 1 & 3 & 4 & -3 \\ 0 & 1 & 3 & -2 \\ 3 & 7 & 6 & -5 \end{bmatrix}$. Is b in the range of the transformation $x \mapsto Ax$? Why or why not?
4. Prove that the transformation $T : \mathbf{R}^m \rightarrow \mathbf{R}^n$ defined by $T(x) = Ax$ satisfy the property: $T(cx) = cT(x)$ for any scalar c .

Answers or Hints are on the second page.

#1: $m = 5$, $n = 7$

#2: $x = (4/9, -37/9, 4/9)$. It is unique. Why?

#3: No. Why? (Hint: The problem is asking if $Ax = b$ is consistent.)

#4: Do as we did in class. Make sure to justify each step.